

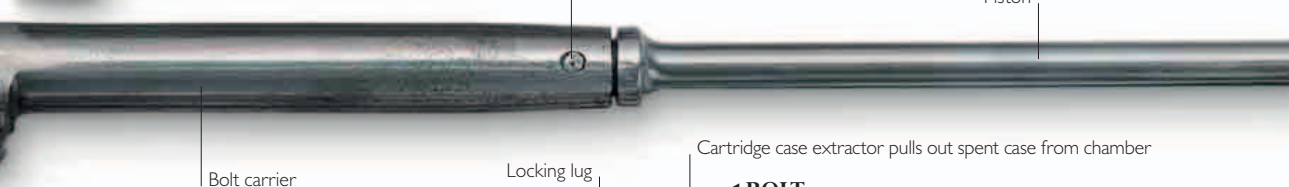
SHOWCASE

AK47

Designed by Mikhail Kalashnikov between 1945 and 1946, the *Avtomat Kalashnikova 47*, or AK47, is the most famous assault rifle in the world. This gun has a gas-operated auto-loading mechanism (see p.305). Its low number of moving parts has helped greatly to reduce its production costs. The AK47 has been adopted by more than 100 armies throughout the world, and its variants are built in more than 30 countries. Amazingly, more than 75 million units have been produced.

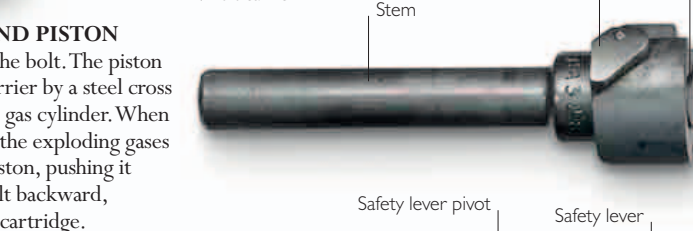
► MAINSPRING

The mainspring fits inside the rear portion of the bolt carrier (the bolt carrier is covered by the action cover). The mainspring's rear is fitted with a ribbed lug that serves as a locking piece for the action cover. When the lug is pushed, the mainspring moves slightly forward so that the cover can be removed.



▲ BOLT CARRIER AND PISTON

The bolt carrier houses the bolt. The piston is attached to the bolt carrier by a steel cross pin. The piston fits in the gas cylinder. When the gun is fired, some of the exploding gases are vented toward the piston, pushing it rearward, driving the bolt backward, and extracting the spent cartridge.



◀ BOLT

Noteworthy for its extremely simple construction, the bolt has a narrow rear stem that fits into the lower part of the bolt carrier. Its head is machined with locking lugs, which slightly retard the rearward movement of the bolt to avoid loss of combustion gas pressure (and consequent bullet velocity) before the bullet leaves the muzzle.



▲ RECEIVER AND BARREL ASSEMBLY

Although initial experiments with pressed steel receivers failed, renewed attempts at producing such receivers after World War II were successful and these became standard production items. The distinctive feature of the barrel is its chromium-plated bore; the chromium minimizes wear and protects against corrosion. The hand guard serves to protect the user's hand from the heat dissipated through the barrel and gas cylinder.

